

Defect Engineering of RuO₂ Aerogel for Efficient Acidic Water Oxidation

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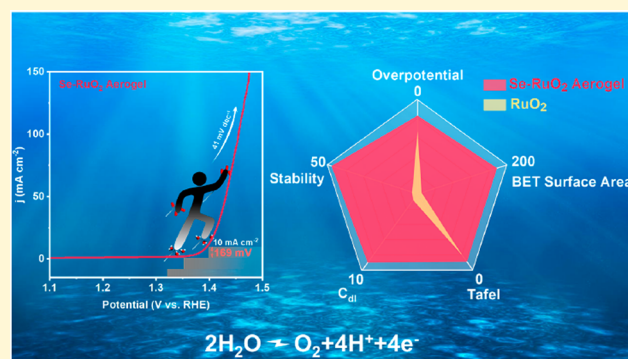


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Supporting Information

ABSTRACT: The development of renewable energy conversion devices heavily relies on the design of high-performance electrocatalysts for water electrolysis systems. In this study, a Se-doped RuO₂ aerogel (Se-RuO₂ aerogel) with abundant defects is prepared as an excellent oxygen evolution reaction (OER) electrocatalyst in acidic media. The Se-RuO₂ aerogel exhibits a remarkably low overpotential of 166 mV at a current density of 10 mA cm⁻² and long-term stability of up to 48 h. Concurrently, detailed *in situ* experiments demonstrate that the Se-RuO₂ aerogel can maintain excellent stability during the OER process, for their reaction path follows a more stable adsorption evolution mechanism. Therefore, it can operate for 100 h at a current density of 100 mA cm⁻² when assembled as an anode catalyst in a polymer electrolyte membrane (PEM) electrolyzer. This work provides a new vision for the design of high-performance electrocatalysts based on aerogel with defect engineering.



With the excessive utilization of fossil fuels and the escalating greenhouse effect, promoting energy transformation has emerged as a global objective.^{1–4} The volatility and inertness of renewable energy sources, such as wind and solar energy, present a formidable challenge to energy utilization. Electrochemical water splitting, as a technology for producing hydrogen which can be stored and utilized to maintain balance between energy supply and demand, is extensively studied.^{5–8} Among the various electrolytic water technologies, polymer electrolyte membrane water electrolyzers (PEMWEs) have received considerable attention due to the fast response to fluctuating electricity, higher current densities, and compact stack designs.^{9–11} However, the anodic oxygen evolution reaction (OER), as the key rate-limiting step of PEMWE, involves a complex four-electron transfer process and thus displays slow reaction kinetics and high overpotential.^{12–16} Moreover, various intermediates like the formation of *O and *OOH have presented significant long-term challenges in devising efficient water splitting catalysts.¹⁷ Consequently, there is an urgent need to design anode catalysts with exceptional OER performance.

Presently, OER catalysts compatible with PEMWE technology predominantly depend on iridium (Ir) and ruthenium (Ru)-based materials.^{18,19} Ruthenium dioxide (RuO₂) is widely recognized as an acidic OER electrocatalyst due to its

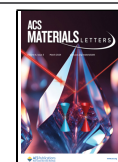
remarkable activity and relatively abundant reserves. However, the irreversible conversion of RuO₂ to soluble RuO₄ species in acidic media severely compromises its stability, which is closely linked to the reaction process.^{20–22} Currently, the prevailing OER mechanisms can be primarily classified into two pathways, namely, the lattice oxygen mechanism (LOM) and the adsorption evolution mechanism (AEM), depending on the origin of oxygen molecules in the reaction products.^{23–25} The AEM pathway is distinct from the LOM pathway in that it does not consume lattice oxygen or produce oxygen vacancies, thereby maintaining the structural integrity of the catalyst during the reaction and resulting in enhanced stability.^{26–28} Clearly, the structural-stability optimization of the catalyst hinges on maximizing the dominance of the AEM.²⁹ To this end, defect engineering, including atomic vacancies (RuO₂/SnO₂), interstitial (C-RuO₂-RuSe), solute atoms (Ru_{0.5}Ir_{0.5}O₂), amorphous/crystalline interface (Am-Ir₁Ru₃O₈), and so on, has received considerable attention.^{30–35} It is worth noting that the presence of defects can disrupt the

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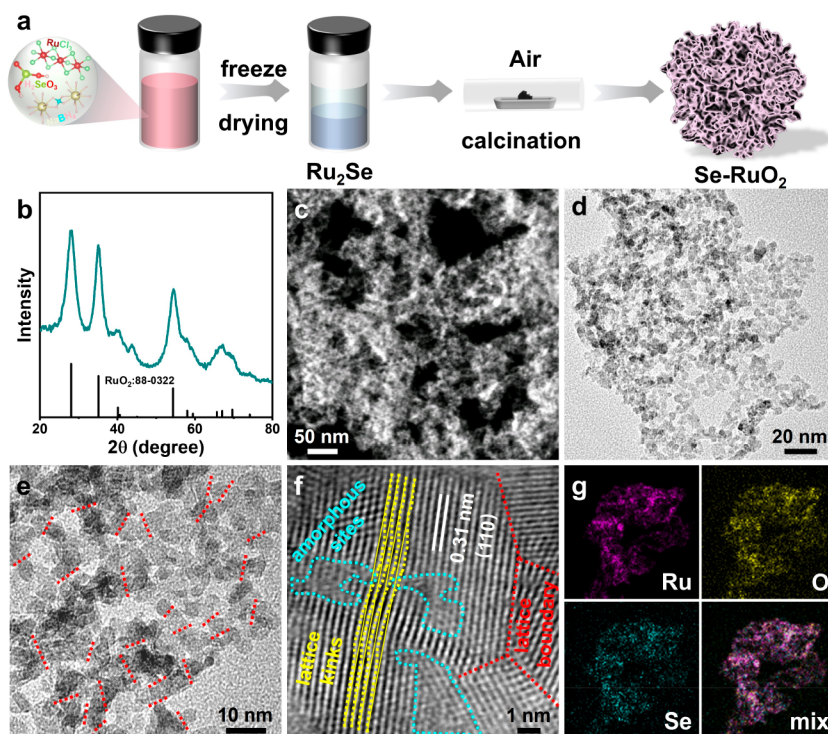


Figure 1. Synthesis and characterization of Se-RuO₂ aerogel. (a) Schematic illustration of the synthetic process for Se-RuO₂ aerogel. (b) XRD pattern, (c) HAADF-STEM image, (d, e) TEM images, (f) HRTEM image, and (g) STEM-EDS elemental mappings of Se-RuO₂ aerogel.

initial structural arrangement, therefore serving as active sites to alter the pathway of the reaction process. For example, Ge et al. successfully employed the MRuO_x solid solution as a structural model for fine-tuning the reaction pathway by regulating the Ru charge, thereby enhancing stability.³⁶ Nevertheless, due to the intricate and challenging nature of defect manipulation, comprehensively investigating the intricate interplay between distinct defects and the inherent catalytic properties of the materials remains a formidable challenge. Recently, metallic aerogel has generated a great deal of interest among researchers, owing to their large internal surface area, high porosity enabling faster mass transfer, and three-dimensional (3D) self-supporting network.^{37–39} Hu et al. had successfully prepared all eight precious metal aerogels (Au, Ag, Pd, Pt, Ru, Rh, Os, Ir) for superior OER and ethanol oxidation reaction (EOR).⁴⁰ Therefore, combining defect engineering and metallic aerogels may be an attractive method to alter the catalytic path of acidic OER based on Ru-based materials.

Herein, the Se-doped RuO₂ aerogel (Se-RuO₂ aerogel) with rich defects was successfully prepared and used as an efficient OER electrocatalyst in acidic conditions. The PEMWE device with the prepared Se-RuO₂ aerogel as catalyst can operate stably for 100 h at a current density of 100 mA cm⁻². Concurrently, *in situ* experiments demonstrated that during acidic OER reaction, the catalyst structure of Se-RuO₂ aerogel was well preserved, and their long-term stability results from the dominant AEM pathway.

A gelation strategy with NaBH₄ as a reducing agent was used to synthesize Ru₂Se aerogel, which was, then, transformed into Se-RuO₂ aerogel through a heat treatment process (Figure 1a and Figure S1). In detail, the initial stage involved blending the precursor solution containing NaBH₄ and XC-72R carbon,

giving rise to a carbon-supporting Ru₂Se hydrogel, which can be converted to a Ru₂Se aerogel after freeze-drying. Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images clearly indicated that the Ru₂Se aerogel was anchored on the surface of the XC-72R carbon (Figure S2a–d). Scanning electron microscopy energy dispersive spectrum (STEM-EDS) demonstrates the uniform distributions of Ru and Se elements (Figure S2e). The atomic ratio of Ru/Se was about 64.1:35.9, as determined by SEM-EDS (Figure S3). Subsequently, the Ru₂Se aerogel was further subjected to oxidation at 300 °C for 15 h, resulting in the formation of Se-RuO₂ aerogel. X-ray diffraction (XRD) results revealed that all of the diffraction peaks corresponded to the rutile-structured RuO₂ (PDF#88–0322) (Figure 1b). It is noteworthy that according to SEM images (Figure S4) and SEM-EDS results (Figure S5), the XC-72R carbon almost completely disappeared and abundant cavities is formed, which can be attributed to the volatilization of Se and the combustion of carbon during the heat treatment process.³⁴ The distinctive features of the Se-RuO₂ aerogel, such as the intricate 3D cross-linked nanostructure and abundant interconnected pores, can be clearly visualized by using high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) (Figure 1c). TEM images in Figure 1d and e showed that the cross-networked structure of the Se-RuO₂ aerogel comprised numerous small grains, which is prone to generating a significant number of defects that is beneficial for enhancing the catalytic performance. Therefore, the defects, including amorphous sites (blue areas), kinks (yellow dotted lines), and boundaries (red dotted lines), can be clearly observed by high-resolution TEM (HRTEM) (Figure 1f). Additionally, the interplanar spacing of 0.31 nm corresponds to the (110) planes of rutile-structured RuO₂, being consistent with the result of

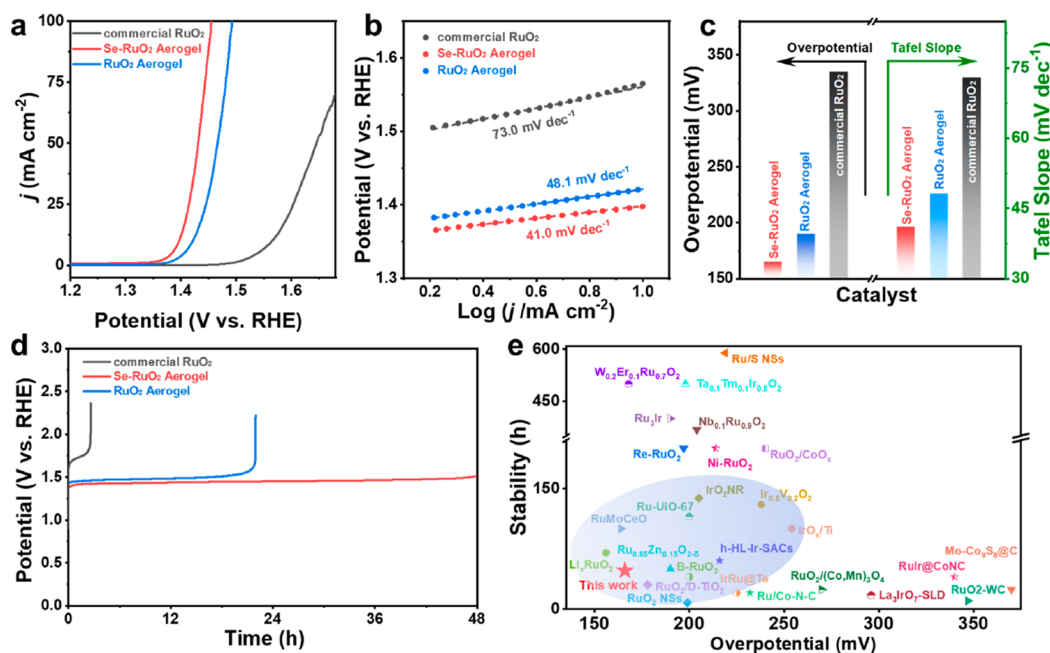


Figure 2. Electrochemical properties of various electrocatalysts. (a) OER polarization curves and (b) corresponding Tafel plots of Se-RuO₂ aerogel, RuO₂ aerogel, and commercial RuO₂. (c) Histogram of overpotentials at 10 mA cm⁻² and Tafel slopes of various catalysts. (d) Chronopotentiometry tests of Se-RuO₂ aerogel, RuO₂ aerogel, and commercial RuO₂ for OER in 0.5 M H₂SO₄ at 10 mA cm⁻². (e) Comparison of the overpotential and stability of Se-RuO₂ aerogel with recently reported OER electrocatalysts in acidic media.

XRD analysis. The homogeneous dispersion of Ru, O and Se elements was confirmed via STEM-EDS elemental mappings, demonstrating the successful synthesis of the Se-RuO₂ aerogel (Figure 1g). For comparison, RuO₂ aerogel was prepared under the same conditions except for the absence of a Se element. The presence of XC-72R carbon further proved that the volatilization of Se can accelerate the combustion of carbon, resulting in the formation of intricate 3D cross-linked nanostructures and abundant interconnected pores (Figures S6 and S7).

Subsequently, the OER performances of the Se-RuO₂ aerogel and RuO₂ aerogel were assessed by a standard three-electrode setup in 0.5 M H₂SO₄, with commercial RuO₂ serving as a reference. The linear sweep voltammograms (LSV) curves are shown in Figure 2a, where the Se-RuO₂ aerogel delivered an excellent electrocatalytic activity with an overpotential of only 166 mV at 10 mA cm⁻², which outdistanced those of the RuO₂ aerogel (210 mV) and commercial RuO₂ (330 mV). In addition, Tafel plots were obtained based on the LSV curves of different catalysts, where the Se-RuO₂ aerogel showed a lower Tafel slope of 41.0 mV dec⁻¹, outperforming the RuO₂ aerogel (48.1 mV dec⁻¹) and commercial RuO₂ (73.0 mV dec⁻¹), which implied faster OER kinetics for the Se-RuO₂ aerogel (Figure 2b). To assess the OER activity, the overpotentials and Tafel slopes of all catalysts are compiled and compared in Figure 2c. Obviously, the OER performance of the Se-RuO₂ aerogel was better than those of the RuO₂ aerogel and commercial RuO₂. The double layer capacitance (C_{dl}) was measured at various scan rates from 10 to 50 mV s⁻¹ to evaluate the electrochemically active surface area (ESCA) (Figure S8). The Se-RuO₂ aerogel exhibited a higher C_{dl} value of 9.0 mF cm⁻² than the RuO₂ aerogel (4.5 mF cm⁻²) and commercial RuO₂ (0.8 mF cm⁻²), implying that the presence of defects will lead to more exposed active sites.

In addition, electrochemical impedance spectroscopy (EIS) was executed to investigate the charge transfer kinetics of the catalyst during the OER process (Figure S9). The charge-transfer resistance (R_{ct}) for Se-RuO₂ aerogel was much smaller than that of commercial RuO₂, indicating that the intricate 3D cross-linked nanostructure and abundant interconnected pores were beneficial to accelerating the charge transfer kinetics during the OER process. Moreover, the chronopotentiometry test was evaluated in 0.5 M H₂SO₄ at 10 mA cm⁻² to assess the stability (Figure 2d). Apparently, the Se-RuO₂ aerogel maintained the majority of their catalytic potential even after 48 h, whereas the activity of RuO₂ aerogel and commercial RuO₂ experienced a significant decline.

Additionally, the influence of temperature on the catalytic performance of Se-RuO₂ aerogel during the heat treatment process was also explored. Both higher (350 °C) and lower (250 °C) synthetic temperature resulted in inferior OER activity and poor stability, demonstrating that the Se-RuO₂ aerogel was extremely sensitive to the annealing temperature (Figures S10–12). Moreover, the shortening or extending the heat treatment time will also affect the performance of the OER, particularly on the stability (Figures S13–15). In Figure 2e and Table S1, the performances of the acidic OER at current density of 10 mA cm⁻² and stability of the Se-RuO₂ aerogel were compared with those of previously reported catalysts, confirming that the stability of Se-RuO₂ aerogel was significantly enhanced while maintaining high activity. In addition, the morphology of the Se-RuO₂ aerogel was largely maintained after electrocatalysis (Figure S16). The XRD pattern of Se-RuO₂ aerogel after OER test did not change obviously, demonstrating the excellent structural stability during the acidic OER process (Figure S17). The Ru 3p binding energy of the Se-RuO₂ aerogel was predominantly maintained, suggesting the absence of noteworthy chemical

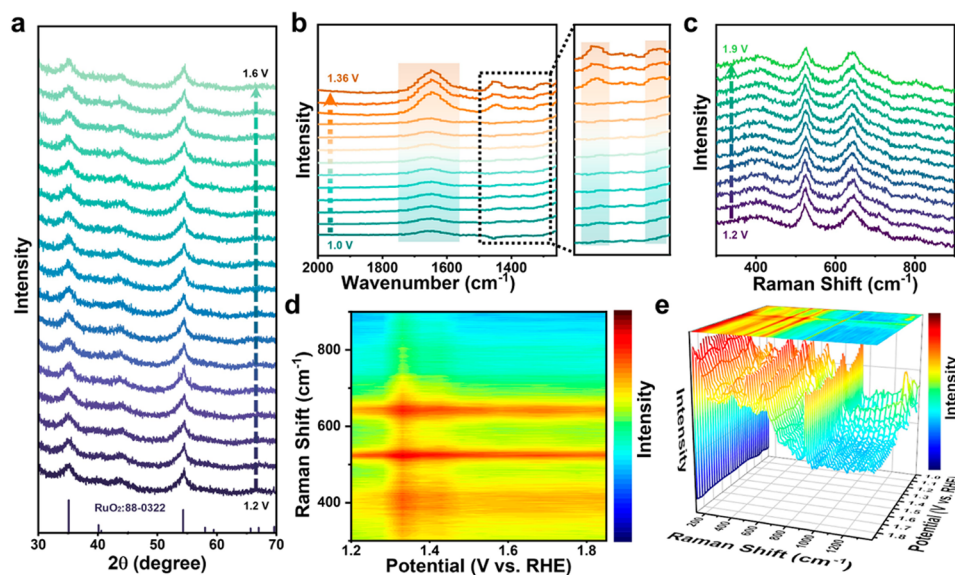


Figure 3. *In situ* measurements. (a) *In situ* XRD patterns and (b) *in situ* ATR-IR spectra recorded during the multipotential steps of Se-RuO₂ aerogel. (c) *In situ* Raman spectrum of Se-RuO₂ aerogel and corresponding (d) contour plots and (e) 3D colormap surface with projection of *in situ* Raman spectrum of Se-RuO₂ aerogel.

and structural alterations in the Ru during the OER measurements (Figure S18). The O 1s XPS spectrum discloses that the O_V/O_L ratio of Se-RuO₂ aerogel remained nearly constant, substantiating that the lattice oxygen within Se-RuO₂ aerogel did not participate in the OER process.

To verify the stability of the Se-RuO₂ aerogel, *in situ* XRD technique was utilized to monitor the real status of the catalysts during the OER process. The X-ray diffraction peak intensity of the Se-RuO₂ aerogel did not decrease significantly when the potential was as high as 1.6 V, showing excellent structural stability within a wide potential range (Figure 3a). In addition, no significant change in the diffraction peak intensity was observed even after 60 min of continuous operation in an acidic environment (Figure S19). Subsequently, *in situ* attenuated total reflection infrared (ATR-IR) measurement was measured to explore the inherent pathway of OER. As shown in Figure 3b, distinct peaks were observed at ~1644, ~1450, and ~1280 cm⁻¹ when the potential reached 1.3 V, corresponding to the HOH bending (δ_{HOH}) mode of water, the O–O stretching mode of weakly adsorbed molecular O₂ (O₂, ad), and the O–O bending mode of surface adsorbed superoxide (OOH_{ad}), respectively.^{41,42} The discernible vibrations of the OOH_{ad} suggested that the reaction on the Se-RuO₂ aerogel followed the AEM pathway, guaranteeing the long-term operation stability. The surface structure dynamics during manipulation of the OER conditions were monitored through *in situ* electrochemical Raman measurements. The vibrational peaks of the Ru–O band were observed in the 500–700 cm⁻¹ range (Figure 3c). The utilization of contour mapping and 3D colormaps, coupled with projection onto *in situ* Raman spectroscopy, facilitates a more vivid depiction of the Raman signal intensity with wave numbers varying (Figure 3d, e). The intensity of the Ru–O characteristic peaks remained relatively stable as the voltage increased, thereby further demonstrating that the OER reaction occurs via the AEM pathway.

To better understand the structure-performance relationship in depth, X-ray photoelectron spectroscopy (XPS) measurements were conducted to comprehend the chemical states and

electron structure of the as-prepared catalysts. For Se-RuO₂ aerogel, two peaks at binding energy values of 463.32 and 485.58 eV can be assigned to the 3p_{3/2} and 3p_{1/2} orbitals of Ru⁴⁺ species, respectively (Figure 4a). The other two peaks (466.54 and 488.69 eV) were attributed to the 3p_{3/2} and 3p_{1/2} orbitals of the Ru³⁺ species, respectively. Notably, the Ru 3p XPS peaks of Se-RuO₂ aerogel shifted toward higher binding energy than those of RuO₂ aerogel, indicating the increase in the Ru valence states. The slight increase of Ru valence state was beneficial to ensuring catalytic stability without sacrificing catalytic activity when the AEM pathway is completely dominant in OER process.³⁶ The O 1s XPS spectra for the Se-RuO₂ aerogel can be assigned to lattice oxygen (O_L), oxygen vacancy (O_V), and surface adsorbed water (O_w), which were located at 530.06, 530.96, and 532.89 eV, respectively (Figure 4b). Obviously, the O_V of the Se-RuO₂ aerogel was significantly enhanced compared to the RuO₂ aerogel, which can be attributed to the presence of abundant defect sites. The Se 3d spectrum of Se-RuO₂ aerogel is shown in Figure 4c, where the peaks located at 60.25 and 58.86 eV can be ascribed to Se⁴⁺ species, proving that the element Se existed in the form of an oxidation state in the Se-RuO₂ aerogel. To experimentally validate the AEM pathway, differential electrochemical mass spectrometry (DEMS) tests were conducted using PM-DEMS coupled with a flow cell (Figure 4d). Subsequently, we assessed O₂ evolution during OER by analyzing isotopic signals in H₂¹⁶O-0.5 M H₂SO₄. As depicted in Figure 4e, distinct ³²O₂ and ³⁴O₂ signal can be observed; however, there is only a flat line for ³⁶O₂, indicating the absence of ³⁶O₂ products. Concurrently, with a progressive escalation in the OER current, a corresponding increase in the ³²O₂ (¹⁶O + ¹⁶O) signal can be observed (Figure 4f). Furthermore, the signal ratio of ³⁴O₂:³²O₂ remained relatively constant within the OER potential range and was comparable to that of the natural abundance of ¹⁸O in deionized water ~2 at. %, suggesting that ³⁴O₂ originated from background ¹⁸O rather than from the lattice (Figure 4g).⁴³ Therefore, a typical AEM pathway is described in Figure 4h: the adsorbed O at the active sites initially interacts with H₂O in solution to form

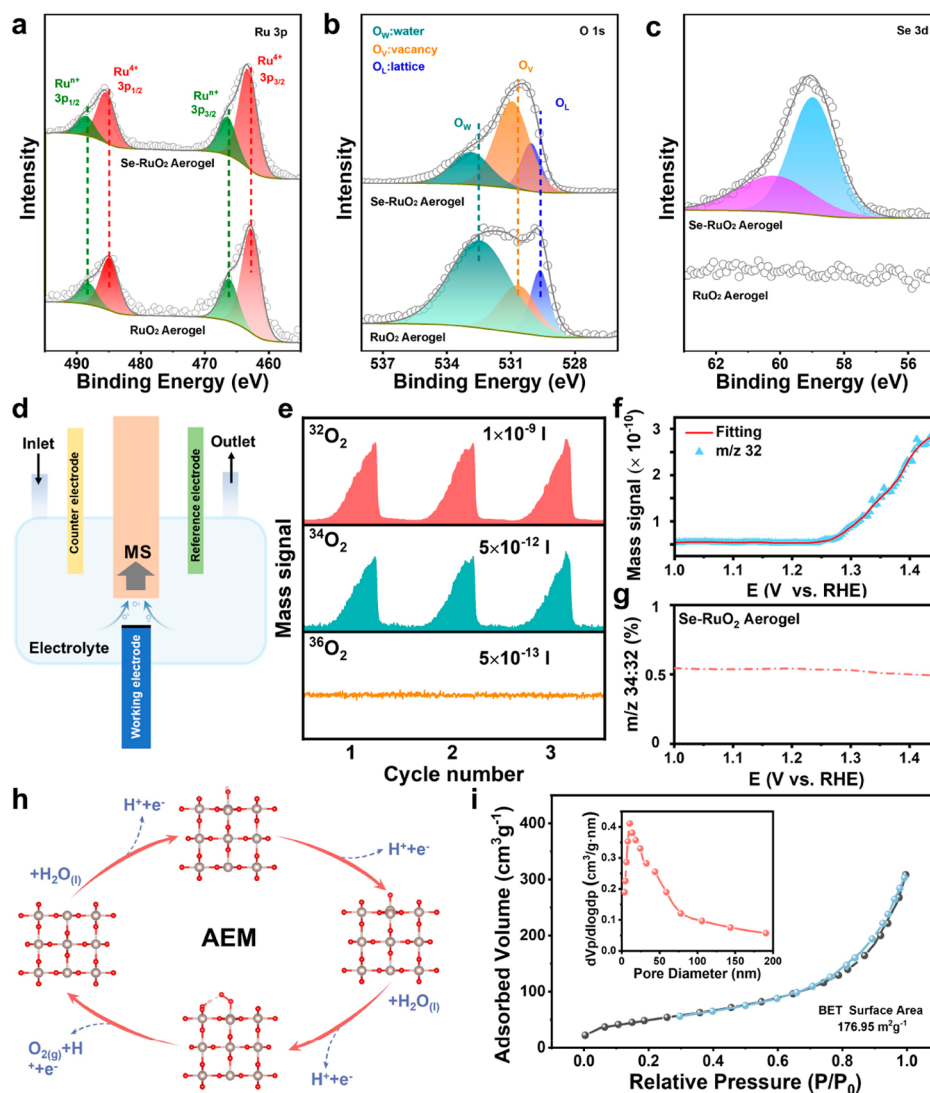


Figure 4. Electronic structure characterization and mechanism analysis. (a) Ru 3p, (b) O 1s, and (c) Se 3d XPS spectra of Se-RuO₂ aerogel and RuO₂ aerogel. (d) Schematic illustration of the DEMS device used. (e) DEMS signals of ³²O₂ (¹⁶O + ¹⁶O), ³⁴O₂ (¹⁶O + ¹⁸O), and ³⁶O₂ (¹⁸O + ¹⁸O) from the gaseous products for ¹⁸O-labeled Se-RuO₂ aerogel. (f) DEMS signals of ³²O₂ collected within the LSV range. (g) Ratio of ³⁴O₂:³²O₂ within the LSV range. (h) The diagram of AEM pathways of RuO₂. (i) N₂ adsorption-desorption isotherm and pore size distribution of Se-RuO₂ aerogel.

*OOH, which subsequently undergoes oxidation to yield oxygen.⁴⁴ Furthermore, the surface area and porosity features were analyzed with the N₂ physisorption isotherm. The surface area of Se-RuO₂ aerogel was 176.95 m² g⁻¹ according to the Brunauer-Emmett-Teller (BET) model, surpassing that of the RuO₂ aerogel (141.36 m² g⁻¹) (Figure 4i and Figure S20). In addition, the concurrently exhibited an almost 20-fold increase compared to the surface area of commercial RuO₂, recorded at 8.87 m² g⁻¹ (Figure S21). The wide diameter distribution from mesopore (<50 nm) to macropore (50–190 nm) can be attributed to the 3D cross-linked nanostructure and abundant interconnected pores, aligning with the observations in the TEM image. The presence of mesoporous and macroporous structures can facilitate good dispersion of catalytic species and effectively hinder the overgrowth of intermediates on the surface of the catalyst, thus stabilizing the exposed active sites.^{45,46} The current was normalized by the BET surface area to portray intrinsic activity, where the Se-

RuO₂ aerogel exhibited a diminished overpotential even when attaining the same specific current density (Figure S22).

Considering the excellent OER activity and durability of Se-RuO₂ aerogel in acidic medium, ultimately, to investigate the potential practical applications of Se-RuO₂ aerogel for water electrolysis, a PEMWE was meticulously constructed. The core membrane electrode assembly (MEA) of PEMWE utilized a catalyst-coated membrane (CCM) configuration with a commendable catalytic layer-membrane interface (Figure 5a). The *I*–*V* curves (without *iR* compensation) of the Pt/C || PEM || Se-RuO₂ aerogel cell provided robust evidence for the excellent electrocatalytic water splitting activity of the Se-RuO₂ aerogel (Figure 5b, c). At an operating temperature of 80 °C, the Se-RuO₂ aerogel-based PEMWE cell demonstrated a commendable performance, achieving current densities of 100 and 500 mA cm⁻² at remarkably low voltages of 1.45 and 1.64 V, respectively (Figure 5d). Subsequently, an evaluation of the stability of the Se-RuO₂ aerogel-based PEMWE cell was conducted and is illustrated in Figure 5e. The Se-RuO₂ aerogel

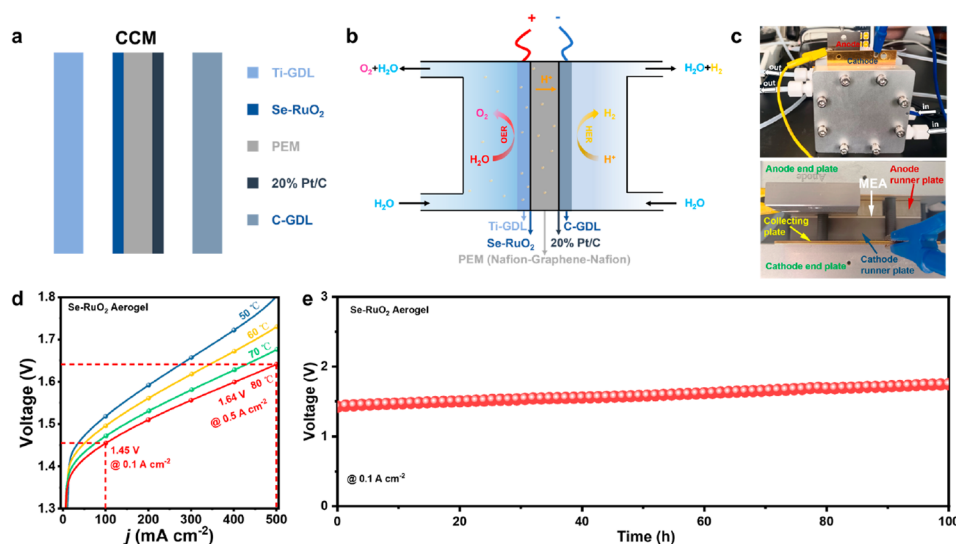


Figure 5. PEMWE cell performance of Se-RuO₂ aerogel. (a) and (b) Schematic diagram of catalyst coating membrane (CCM) for membrane electrodes (AEM). (c) Photograph of the PEM water electrolyzer. (d) Polarization curves of PEMWEs using Se-RuO₂ aerogel as anodes at 50–80 °C with Nafion 115 membrane without *iR* compensation. (e) Chronopotentiometry testing of Se-RuO₂ aerogel and at 100 mA cm^{−2} in the PEMWE.

exhibited excellent PEMWE durability with negligible performance degradation after 100 h. The great activity and stability demonstrated the immense potential of the Se-RuO₂ aerogel for future industrial implementations.

In summary, the Se-RuO₂ aerogel has been successfully prepared by a gelation strategy and heat treatment process. The abundant defect and interconnected pores can be utilized to enhance the exposure of active sites, thereby augmenting both the OER activity and stability in acidic environments. The judicious *in situ* measurements and DEMS substantiated that the Se-RuO₂ aerogel achieved the OER reaction via the AEM pathway and that the catalyst exhibited robust stability during the OER process. The PEMWE achieved a current density of 500 mA cm^{−2} at low battery voltage of 1.64 V and can be operated stably for 100 h at 100 mA cm^{−2}. This study provides a feasible solution for the practical implementation of Ru-based catalysts in commercial PEMWE.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsmaterialslett.3c01414>.

Experimental Procedures; Figures S1–S22, including SEM images, SEM-EDS spectra, XRD patterns, cyclic voltammograms, Nyquist plots, OER polarization curves, TEM images, XPS spectra, and N₂ adsorption–desorption isotherms; Table S1 (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Zhou, X.; Chu, S.; Jin, Z.; Hu, K.; Liu, P.; Qiu, H.-J.; Lin, X. Revealing the synergistic enhancement effect of dual metal RuFe (Co) sites for bifunctional oxygen catalysis. *ACS Materials Lett.* **2023**, *5*, 1656–1664.
- (2) Wei, X.; Cao, S.; Xu, H.; Jiang, C.; Wang, Z.; Ouyang, Y.; Lu, X.; Dai, F.; Sun, D. Novel two-dimensional metal organic frameworks: high-performance bifunctional electrocatalysts for OER/ORR. *ACS Materials Lett.* **2022**, *4*, 1991–1998.
- (3) Meng, Z.; Qiu, Z.; Shi, Y.; Wang, S.; Zhang, G.; Pi, Y.; Pang, H. Micro/nano metal–organic frameworks meet energy chemistry: A review of materials synthesis and applications. *eScience* **2023**, *3*, No. 100092.
- (4) Wang, L.; Xu, Z.; Kuo, C.; Peng, J.; Hu, F.; Li, L.; Chen, H.; Wang, J.; Peng, S. Stabilizing low-valence single atoms by constructing metalloid tungsten carbide supports for efficient hydrogen oxidation and evolution. *Angew. Chem., Int. Ed.* **2023**, *135*, No. e202311937.
- (5) He, X.; Han, X.; Zhou, X.; Chen, J.; Wang, J.; Chen, Y.; Yu, L.; Zhang, N.; Li, J.; Wang, S.; Jin, H. Electronic modulation with Pt-incorporated NiFe layered double hydroxide for ultrastable overall water splitting at 1000 mA cm⁻². *Appl. Catal. B-Environ.* **2023**, *331*, No. 122683.
- (6) Chen, D.; Qiao, M.; Lu, Y.; Hao, L.; Liu, D.; Dong, C.; Li, Y.; Wang, S. Preferential cation vacancies in perovskite hydroxide for oxygen evolution reaction. *Angew. Chem., Int. Ed.* **2018**, *57*, 8691.
- (7) Zhou, Y.; Fan, H. J. Progress and challenge of amorphous catalysts for electrochemical water splitting. *ACS Materials Lett.* **2021**, *3*, 136–147.
- (8) Zhu, J.; Lu, R.; Shi, W.; Gong, L.; Chen, D.; Wang, P.; Chen, L.; Wu, J.; Mu, S.; Zhao, Y. Epitaxially grown Ru clusters–nickel nitride heterostructure advances water electrolysis kinetics in alkaline and seawater media. *Energy Environ. Mater.* **2023**, *6*, No. e12318.
- (9) Deng, L.; Hung, S.; Lin, Z.; Zhang, Y.; Zhang, C.; Hao, Y.; Liu, S.; Kuo, C.; Chen, H.; Peng, J.; et al. Valence oscillation of Ru active sites for efficient and robust acidic water oxidation. *Adv. Mater.* **2023**, *35*, No. 2305939.
- (10) Li, L.; Zhou, J.; Wang, X.; Gracia, J.; Valvidares, M.; Ke, J.; Fang, M.; Shen, C.; Chen, J.; Chang, Y.; et al. Spin-polarization strategy for enhanced acidic oxygen evolution activity. *Adv. Mater.* **2023**, *35*, No. 2302966.
- (11) Chen, Z.; Guo, L.; Pan, L.; Yan, T.; He, Z.; Li, Y.; Shi, C.; Huang, Z.; Zhang, X.; Zou, J. Advances in oxygen evolution electrocatalysts for proton exchange membrane water electrolyzers. *Adv. Energy Mater.* **2022**, *12*, No. 2103670.
- (12) Xiao, Z.; Huang, Y.; Dong, C.; Xie, C.; Liu, Z.; Du, S.; Chen, W.; Yan, D.; Tao, L.; Shu, Z.; Zhang, G.; Duan, H.; Wang, Y.; Zou, Y.; Chen, R.; Wang, S. Operando identification of the dynamic behavior of oxygen vacancy-rich Co₃O₄ for oxygen evolution reaction. *J. Am. Chem. Soc.* **2020**, *142*, 12087–12095.
- (13) Jia, Z.; Yuan, Y.; Zhang, Y.; Lyu, X.; Liu, C.; Yang, X.; Bai, Z.; Wang, H.; Yang, L. Optimizing 3d spin polarization of CoOOH by in situ Mo doping for efficient oxygen evolution reaction. *Carbon Energy* **2023**, No. e418.
- (14) Zhao, S.; Wang, Y.; Hao, Y.; Yin, L.; Kuo, C.; Chen, H.; Li, L.; Peng, S. Lewis acid driving asymmetric interfacial electron distribution to stabilize active species for efficient neutral water oxidation. *Adv. Mater.* **2023**, No. 2308925.
- (15) Liu, L.; Ji, Y.; You, W.; Liu, S.; Shao, Q.; Kong, Q.; Hu, Z.; Tao, H.; Bu, L.; Huang, X. Trace lattice S inserted RuO₂ flexible nanosheets for efficient and long-term acidic oxygen evolution catalysis. *Small* **2023**, *19*, No. 2208202.
- (16) Zhang, L.; Zhu, J.; Li, X.; Mu, S.; Verpoort, F.; Xue, J.; Kou, Z.; Wang, J. Nurturing the marriages of single atoms with atomic clusters and nanoparticles for better heterogeneous electrocatalysis. *Interdisciplinary Materials* **2022**, *1*, 51–87.
- (17) Feng, T.; Chen, W.; Xue, J.; Cao, F.; Chen, Z.; Ye, J.; Xiao, C.; Lu, H. Spin polarization of chiral amorphous Fe–Ni electrocatalysts enabling efficient electrochemical oxygen evolution. *Adv. Funct. Mater.* **2023**, *33*, No. 2215051.
- (18) Liu, S.; Tan, H.; Huang, Y.; Zhang, Q.; Lin, H.; Li, L.; Hu, Z.; Huang, W.; Pao, C.; Lee, J.; et al. Structurally-distorted RuIr-based nanoframes for long-duration oxygen evolution catalysis. *Adv. Mater.* **2023**, *35*, No. 2305659.
- (19) Zhang, Y.; Chen, M.; Guo, P.; Du, Y.; Song, B.; Wang, X.; Jiang, Z.; Xu, P. Magnetic field-enhanced water splitting enabled by bifunctional molybdenum-doped nickel sulfide on nickel foam. *Carbon Energy* **2023**, No. e351.
- (20) Hu, C.; Xu, J.; Tan, Y.; Huang, X. Recent advances of ruthenium-based electrocatalysts for hydrogen energy. *Trends Chem.* **2023**, *5*, 225–239.
- (21) Deka, N.; Jones, T. E.; Falling, L. J.; Sandoval-Diaz, L.-E.; Lunkenbein, T.; Velasco-Velez, J.-J.; Chan, T.-S.; Chuang, C.-H.; Knop-Gericke, A.; Mom, R. V. On the operando structure of ruthenium oxides during the oxygen evolution reaction in acidic media. *ACS Catal.* **2023**, *13*, 7488–7498.
- (22) Zhu, W.; Yao, F.; Cheng, K.; Zhao, M.; Yang, C.; Dong, C.; Hong, Q.; Jiang, Q.; Wang, Z.; Liang, H. Direct dioxygen radical coupling driven by octahedral ruthenium–oxygen–cobalt collaborative coordination for acidic oxygen evolution reaction. *J. Am. Chem. Soc.* **2023**, *145*, 17995–18006.
- (23) Yoon, H.; Ju, B.; Kim, D. Perspectives on the development of highly active, stable, and cost-effective OER electrocatalysts in acid. *Battery Energy* **2023**, *2*, No. 20230017.
- (24) Chen, F.; Wu, Z.; Adler, Z.; Wang, H. Stability challenges of electrocatalytic oxygen evolution reaction: from mechanistic understanding to reactor design. *Joule* **2021**, *5*, 1704–1731.
- (25) Hao, Y.; Hung, S.; Zeng, W.; Wang, Y.; Zhang, C.; Kuo, C.; Wang, L.; Zhao, S.; Zhang, Y.; Chen, H.; Peng, S. Switching the oxygen evolution mechanism on atomically dispersed Ru for enhanced acidic reaction kinetics. *J. Am. Chem. Soc.* **2023**, *145* (43), 23659–23669.
- (26) He, J.; Chen, W.; Gao, H.; Chen, Y.; Zhou, L.; Zou, Y.; Chen, R.; Tao, L.; Lu, X.; Wang, S. Tuning hydrogen binding modes within RuO₂ lattice by proton and electron Co-doping for active and stable acidic oxygen evolution. *Chem. Catal.* **2022**, *2*, 578–594.
- (27) Song, J.; Wei, C.; Huang, Z.; Liu, C.; Zeng, L.; Wang, X.; Xu, Z. A review on fundamentals for designing oxygen evolution electrocatalysts. *Chem. Soc. Rev.* **2020**, *49*, 2196–2214.
- (28) Wang, Y.; Lei, X.; Zhang, B.; Bai, B.; Das, P.; Azam, T.; Xiao, J.; Wu, Z. Breaking the Ru–O–Ru symmetry of a RuO₂ catalyst for sustainable acidic water oxidation. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e20231690.
- (29) Reier, T.; Nong, H. N.; Teschner, D.; Schlögl, R.; Strasser, P. Electrocatalytic oxygen evolution reaction in acidic environments—reaction mechanisms and catalysts. *Adv. Energy Mater.* **2017**, *7*, No. 1601275.
- (30) Huang, B.; Cui, Y.; Liu, X.; Zheng, C.; Wang, H.; Guan, L. Dense-packed RuO₂ nanorods with in situ generated metal vacancies loaded on SnO₂ nanocubes for proton exchange membrane water

electrolyzer with ultra-low noble metal loading. *Small* **2023**, *19*, No. 2301516.

(31) Zhu, W.; Song, X.; Liao, F.; Huang, H.; Shao, Q.; Feng, K.; Zhou, Y.; Ma, M.; Wu, J.; Yang, H.; et al. Stable and oxidative charged Ru enhance the acidic oxygen evolution reaction activity in two-dimensional ruthenium-iridium oxide. *Nat. Commun.* **2023**, *14*, 5365.

(32) Mao, Q.; Mu, X.; Wang, W.; Deng, K.; Yu, H.; Wang, Z.; Xu, Y.; Wang, L.; Wang, H. Atomically dispersed Cu coordinated Rh metallene arrays for simultaneously electrochemical aniline synthesis and biomass upgrading. *Nat. Commun.* **2023**, *14*, 5679.

(33) Zhao, S.; Hu, F.; Yin, L.; Li, L.; Peng, S. Manipulating electron re-distribution induced by asymmetric coordination for electrocatalytic water oxidation at a high current density. *Sci. Bull.* **2023**, *68*, 1389–1398.

(34) Wang, J.; Cheng, C.; Yuan, Q.; Yang, H.; Meng, F.; Zhang, Q.; Gu, L.; Cao, J.; Li, L.; Haw, S.; Shao, Q.; Zhang, L.; Cheng, T.; Jiao, F.; Huang, X. Exceptionally active and stable RuO₂ with interstitial carbon for water oxidation in acid. *Chem.* **2022**, *8*, 1673–1687.

(35) Li, L.; Cheng, Z.; Su, J.; Song, B.; Yu, H.; Ji, Y.; Shao, Q.; Lu, J. One-dimensional amorphous porous iridium-ruthenium oxide for efficient acidic oxygen evolution reaction. *J. Mater. Chem. A* **2023**, *11*, No. 25268.

(36) Shi, Z.; Li, J.; Wang, Y.; Liu, S.; Zhu, J.; Yang, J.; Wang, X.; Ni, J.; Jiang, Z.; Zhang, L.; et al. Customized reaction route for ruthenium oxide towards stabilized water oxidation in high-performance PEM electrolyzers. *Nat. Commun.* **2023**, *14*, 843.

(37) Huang, H.; Lai, F.; Fu, H.; Chen, Y.; Li, H.; He, F.; Wang, Z.; Zhang, N.; Bai, S.; Liu, T. Spin-engineered Cu-Ni metallic aerogels for enhanced ethylamine electrosynthesis from acetonitrile. *J. Mater. Chem. A* **2023**, *11*, 2210–2217.

(38) Li, H.; Huang, H.; Chen, Y.; Lai, F.; Fu, H.; Zhang, L.; Zhang, N.; Bai, S.; Liu, T. High-entropy alloy aerogels: a new platform for carbon dioxide reduction. *Adv. Mater.* **2023**, *35*, No. 2209242.

(39) Fu, H.; Wang, J.; Chen, Y.; Huang, H.; Wang, Y.; Li, H.; Lai, F.; Zhang, L.; Zhang, N.; Chen, J.; Liu, T. Pd₃Pb bimetallic aerogels for anti-poisoned oxygen reduction reaction. *Chem. Eng. J.* **2023**, *470*, No. 144255.

(40) Du, R.; Wang, J.; Wang, Y.; Hübner, R.; Fan, X.; Senkovska, I.; HU, Y.; Kaskel, S.; Eychmüller, A. Unveiling reductant chemistry in fabricating noble metal aerogels for superior oxygen evolution and ethanol oxidation. *Nat. Commun.* **2020**, *11*, 1590.

(41) Ogose, T.; Kasahara, S.; Ikemiya, N.; Hoshi, N.; Einaga, Y.; Nakamura, M. In situ ATR-IR observation of the electrochemical oxidation of a polycrystalline boron-doped diamond electrode in acidic solutions. *J. Phys. Chem. C* **2018**, *122*, 27456–27461.

(42) Nayak, S.; McPherson, I.; Vincent, K. Adsorbed intermediates in oxygen reduction on platinum nanoparticles observed by in situ IR spectroscopy. *Angew. Chem., Int. Ed.* **2018**, *57*, No. 12855.

(43) Stoerzinger, K.; Diaz-Morales, O.; Kolb, M.; Rao, R.; Frydendal, R.; Qiao, L.; Wang, X.; Halck, N.; Rossmeisl, J.; Hansen, H.; Vegge, T.; Stephens, I.; Koper, M.; Shao-Horn, M. Orientation-dependent oxygen evolution on RuO₂ without lattice exchange. *ACS Energy Lett.* **2017**, *2*, 876–881.

(44) Ma, Q.; Mu, S. Acidic oxygen evolution reaction: Mechanism, catalyst classification, and enhancement strategies. *Interdisciplinary Materials* **2023**, *2*, 53–90.

(45) Shi, J. On the synergetic catalytic effect in heterogeneous nanocomposite catalysts. *Chem. Rev.* **2013**, *113*, 2139–2181.

(46) Liu, G.; Yang, Y.; Zhang, X.; Li, H.; Yu, P.; Gao, M.; Yu, S. Porous tellurium-doped ruthenium dioxide nanotubes for enhanced acidic water oxidation. *Small* **2023**, No. 2306914.